

DevOps Practical Assessment – Minimum Requirements

Objective

Evaluate the candidate's practical DevOps skills by deploying an application using their own cloud infrastructure while utilizing a company-provided cPanel account for DNS management.

Candidate Must Have

- AWS Account (Mandatory)
 - GitHub Account
 - Docker Installed
 - Linux/Ubuntu Environment
 - Internet Connectivity
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Assessment Requirements

- Deploy the provided application on an AWS EC2 instance.
- Configure the server with all required dependencies.
- Configure application environment variables.
- Run the application as a service.
- Ensure the application is accessible via the EC2 Public IP.
- Install and configure the Apache HTTP Server.
- Configure Apache Virtual Hosts for the assigned domain/subdomain.
- Configure Apache as a Reverse Proxy (if applicable) to forward requests to the application.
- Verify the application is accessible through the Apache web server.
- Configure DNS using the provided cPanel account.
- Create the required DNS records (A Record/CNAME as applicable).
- Point the provided domain/subdomain to the AWS EC2 instance.
- Verify that the application is accessible using the assigned domain.
- Configure HTTPS using SSL (Let's Encrypt or a valid SSL certificate).
- Create a Dockerfile for the application.
- Build the Docker image successfully.
- Run the application inside a Docker container.
- Expose the required application port.
- Verify that the containerized application is accessible.
- Push the application source code to a GitHub repository.
- Provide basic deployment documentation (README).
- Demonstrate troubleshooting by resolving one deployment or configuration issue introduced during the assessment.

- Explain the deployment architecture and the decisions taken during implementation.
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Deliverables

- GitHub Repository Link
 - AWS EC2 Public IP
 - Working Domain/Subdomain URL
 - Dockerfile
 - Apache Virtual Host Configuration File
 - README with deployment steps
 - Screenshots of successful deployment
 - Docker container verification (`docker ps`)
 - Apache service verification (`systemctl status apache2` or `httpd`)
 - Brief explanation of the deployment architecture
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Evaluation Criteria

- Linux Administration
 - AWS EC2 Configuration
 - Apache HTTP Server Configuration
 - Virtual Host Configuration
 - Reverse Proxy Configuration
 - DNS Configuration
 - SSL Configuration
 - Docker Fundamentals
 - Networking Knowledge
 - Application Deployment
 - Troubleshooting & Debugging
 - Documentation Quality
 - Problem-Solving Approach
 - Communication and Technical Explanation
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Expected Outcome

The candidate should successfully demonstrate the ability to:

- Deploy an application to a cloud environment.
- Configure Apache HTTP Server to host the application.
- Configure Virtual Hosts and Reverse Proxy where required.
- Configure DNS to expose the application through a domain.
- Secure the application using HTTPS.
- Containerize the application using Docker.
- Troubleshoot deployment, networking, Apache, and application-related issues independently.

- Explain the deployment workflow and architecture confidently.